

MATERIAL TRANSFER AGREEMENT FOR HERBARIUM SPECIMENS AND TISSUE COLLECTIONS

Revised Apr 2025

The Missouri Botanical Garden (MBG) releases material from its DNA Bank (consisting of leaf tissue in silica gel kept at -20°C) or herbarium collection for destructive sampling for DNA and other analyses only under specific conditions and in support of appropriate systematic research projects. Samples will not be made available for bioprospecting endeavors, screening for genes of interest in agricultural or applied research, or any other potential commercial application. MBG supports *The Principles on Access to Genetic Resources and Benefit-Sharing* (<https://www.bgci.org/resources/bgci-tools-and-resources/principles-on-access-to-genetic-resources-and-benefit-sharing/>) and requires all collaborators receiving material from its collections to abide by the letter and spirit of the Nagoya Protocol and the Convention on Biological Diversity (<https://www.cbd.int/abs/>).

The investigator must have made a diligent and good faith effort to obtain appropriate material for extraction from other sources (field collections, horticultural sources, other herbaria, etc.). Specimens in the DNA Bank must take priority over specimens of the same taxa in the herbarium for sampling. The DNA Bank should thus be consulted before submitting a destructive sampling request and can be searched here: <http://legacy.tropicos.org/DNASpecimenSearch.aspx>. To offset a portion of the costs of processing, maintaining, and distributing samples from MBG's DNA Bank, a contribution of \$US25.00 per sample is charged. For students who lack adequate funding, a request for a partial or complete waiver may be submitted to the Director of the Herbarium by the student's major professor. Under exceptional circumstances, a waiver may be granted to other researchers. Due to the unknown quality of DNA present in herbarium specimens, there is no charge for these samples, though the researcher may be asked to help cover processing and/or shipping costs for larger requests.

The investigator must have enough experience with the laboratory techniques and the characteristics of their group to insure that there is a reasonable probability of success using the requested material.

As a condition of release for any samples from herbarium material, each applicant agrees to abide by the conditions stated in the paragraphs above and agrees to:

- 1) Obtain approval in advance to pass either material provided by the Garden or extracted DNA to third parties.
- 2) Acknowledge both MBG and each individual collector of material provided in each publication in which data is used, and provide the Garden with reprints from all resultant publications.
- 3) Publish jointly with MBG staff members or their collaborators whenever appropriate.
- 4) Register DNA sequences with GenBank/EMBL accession numbers.
- 5) Acknowledge that the material is provided by the Garden without any warranty of any kind whatsoever.
- 6) Comply with these specific sampling requirements:
 - a. Only the minimum amount of material may be taken, and only from the fragment packets.
 - b. Only one or two specimens per taxon may be sampled and no more than 20 total specimens per project.
 - c. Sampling is restricted to non-type specimens collected since 1970, and a given specimen may not be sampled a second time.

- d. Any sampled specimen must be clearly annotated with what material was removed, by whom, their institution and for what purpose.

Exceptions may be made to this policy in specific instances, but will only be granted after thorough review by the Curator of the Herbarium at the Missouri Botanical Garden.

The Missouri Botanical Garden reserves the right to refuse to provide samples from the herbarium material in its care for any reason, at its sole discretion.

*I certify that I have read and understand the above restrictions and agree that I will conform to all of the regulations of the Missouri Botanical Garden.

Printed Name Wen-Hsi Kuo

Institution name and acronym Missouri Botanical Garden

Project title Conservation management of Quercus arkansana

Signature Wen-Hsi Kuo Date Jun 16, 2026

*Please provide the following information for each specimen to be sampled in a separate list:

Family Name, Taxon Name, Country, Collector & number, Collection Date, Specimen Accession number

A separate MTA and all accompanying documentation must be submitted for each sampling event.

Send this completed MTA, a brief 1-page project description (including methods for processing herbarium samples, the amount of material needed, likelihood of success, and reason that material from other sources could not be acquired), and specimen list to:

Curator of the Herbarium
Missouri Botanical Garden
4344 Shaw Blvd.
St. Louis, Missouri 63166-0299, USA
Email (preferred): herbarium@mobot.org

For MO Internal Use Only:

Transaction No. _____

Curator of the Herbarium

Date